

1. (15 points) An insulated cylinder contains 5.0 liters of air at room temperature (300 K) and atmospheric pressure. (10^5 N/m^2). At one end of the cylinder is a massless piston, whose surface area is 0.01 m^2 . Suppose that you push the piston in very hard, exerting a force of 2500 N. The piston moves 4 mm before it is stopped.

a. (5 points) Calculate the amount of work you did on the system, and the heat added to the gas. Assuming that all of the energy added goes into the gas (not the piston or cylinder walls), by how much does the internal energy of the gas increase?

$$\Delta U = Q + W$$

$$W = F d = 2500 \text{ N} \cdot 4 \times 10^{-3} \text{ m} = 10 \text{ J} \quad PV = nRT$$

$Q = 0$ because the system is insulated

$$\Delta U = W = 10 \text{ J}$$

b. (5 points) Calculate the change in the entropy of the gas once it has again reached equilibrium. (Because the piston displacement is tiny compared with the total volume of gas, you may assume the internal pressure and temperature of the gas are constant in this problem).

$$S = Nk \left[\ln \left(\frac{V}{N} \left(\frac{4\pi m U}{2\pi h^2} \right)^{3/2} \right) + \frac{5}{2} \right]$$

$$\Delta V = 0.01 \text{ m}^2 (-4 \times 10^{-3} \text{ m}) = -4 \times 10^{-5} \text{ m}^3$$

$$PV = NkT$$

$$\frac{PV_i}{T} = Nk$$

$$\Delta S = \frac{P \Delta V}{T} \left[\ln \left(\frac{V_f}{N} \left(\frac{4\pi m U}{2\pi h^2} \right)^{3/2} \right) + \frac{5}{2} \right] - \left[\frac{PV_i}{T} \left[\ln \left(\frac{V_i}{N} \left(\frac{4\pi m U}{2\pi h^2} \right)^{3/2} \right) + \frac{5}{2} \right] \right] \quad (\text{most can be cancelled by log properties})$$

$$\Delta S = \frac{P \Delta V}{T} \ln \left(\frac{V_f}{V_i} \right)$$

$$1 \text{ L} = 10^{-3} \text{ m}^3$$

$$5 \text{ L} = 5 \times 10^{-3} \text{ m}^3$$

$$\Delta S = \frac{P \Delta V}{T} \cdot \frac{5}{2} \ln \frac{V_f}{V_i} = \frac{10^5 \text{ N/m}^2 \cdot (-4 \times 10^{-5} \text{ m}^3)}{300 \text{ K}} \cdot \frac{5}{2} \ln \left(1 + \frac{-4 \times 10^{-5} \text{ m}^3}{5 \times 10^{-3} \text{ m}^3} \right) = \boxed{2.68 \times 10^{-4} \text{ J/K}} \quad \frac{V_f}{V_i} = \frac{\Delta V + V_i}{V_i} = \frac{\Delta V}{V_i} + 1$$

c. (5 points) Is this process reversible or irreversible? Explain. If it is reversible, how could you make it irreversible? Likewise, if it is irreversible, how could you make it reversible?

This process is irreversible because ΔS is positive. Theoretically, we could make this process reversible by doing this process very slowly, letting the system come to equilibrium at every point of change so that ΔS is no longer > 0 .

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2. (15 points) The multiplicity for a cold, large Einstein solid is:

$$\Omega \approx \left(\frac{eN}{q} \right)^q$$

With the system's total energy defined by:

$$U = \epsilon q$$

(a) (5 points) Calculate the temperature of the solid as a function of its total energy.

$$S = k \ln(\Omega) = k \ln \left(\frac{eN}{q} \right)^q$$

$$U = \epsilon q \rightarrow q = \frac{U}{\epsilon}$$

$$S = k \cdot \frac{U}{\epsilon} \ln \left(\frac{eN\epsilon}{U} \right)$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,V} = \frac{\partial}{\partial U} \left[\frac{kU}{\epsilon} \ln \left(\frac{eN\epsilon}{U} \right) \right] = \frac{k}{\epsilon} \ln \left(\frac{eN\epsilon}{U} \right) + \frac{kU}{\epsilon} \cdot -\frac{1}{U}$$

$$\ln(eN\epsilon) - \ln U$$

$$\frac{1}{T} = \frac{k}{\epsilon} (\ln(eN\epsilon) - 1)$$

$$T = \frac{\epsilon}{k (\ln(eN\epsilon) - 1)}$$

(b) (5 points) Calculate the energy of the solid as a function of temperature.

$$\frac{1}{T} = \frac{k}{\epsilon} (\ln(eN\epsilon) - 1) \Rightarrow \ln \left(\frac{eN\epsilon}{U} \right) = \frac{\epsilon}{kT} + 1$$

$$\frac{eN\epsilon}{U} = e^{\frac{\epsilon}{kT} + 1}$$

$$U = \frac{eN\epsilon}{e^{\frac{\epsilon}{kT} + 1}}$$

(c) (5 points) Calculate the constant volume heat capacity of the solid.

$$\begin{aligned}
 C_v &= \left(\frac{\partial U}{\partial T} \right)_{N,V} \\
 U &= \frac{eN\epsilon}{e^{\epsilon/kT} + 1} \\
 &= \frac{2}{\partial T} \left(\frac{eN\epsilon}{e^{\epsilon/kT} + 1} \right) = \frac{2}{\partial T} eN\epsilon (e^{\epsilon/kT} + 1)^{-1} \\
 &= -eN\epsilon (e^{\epsilon/kT} + 1)^{-2} \cdot \frac{\epsilon}{k} T^{-2} e^{\epsilon/kT} \\
 &= \frac{eN\epsilon^2 e^{\epsilon/kT}}{T^2 (e^{\epsilon/kT} + 1)^2}
 \end{aligned}$$

3. (15 points) Calcium carbonate has two common crystalline forms, calcite and aragonite, with the following thermodynamic data:

Form	$\Delta_f H$ (kJ)	$\Delta_f G$ (kJ)	S (J/K)	V (cm ³)
calcite	-1206.9	-1128.8	92.9	36.93
aragonite	-1207.1	-1127.8	88.7	34.15

All data are for one mole of material at 298 K and 1 bar. S and V can be considered independent of temperature for this problem.

$$1 \text{ bar} = 10^5 \text{ N/m}^2$$

a. (5 points) Which form is stable at the earth's surface and why?

Calcite is stable at earth's surface because $\Delta_f G$ is lowest for calcite.

b. (5 points) Assume the temperature is kept at room temperature. Calculate the pressure at which the phase not stable in part a should become stable.

$$\begin{aligned}
 H &= U + PV \\
 G &= U + PV - TS = H - TS \\
 dG &= -SdT + VdP + \mu dN \\
 \Delta G_c &= \Delta G_a \\
 -1128.8 \text{ kJ} &= dG_a - SdT + V_A dP + \mu dN \\
 -1128.8 \text{ kJ} &= -SdT + V_A dP + \mu dN \\
 T &\text{ is constant} \\
 N &\text{ is constant} \\
 -1128.8 \text{ kJ} &= -1127.8 + V_A dP \\
 1 \text{ kJ} &= V_A dP \\
 dP_A &= \frac{-1000 \text{ J}}{34.15 \times 10^{-6} \text{ m}^3} \\
 10 & \dots \dots \dots -797 \text{ bar}
 \end{aligned}$$